

Bulk Density, Saturation Water Content and Rate of Wetting of Soil Aggregates¹

F. A. GUMBS AND B. P. WARKENTIN²

ABSTRACT

Bulk density, degree of saturation, volume increase on wetting, and rate of wetting were measured for aggregates and clods of Ormstown silty clay loam (Humaquept) of sizes from 5 cm to 0.036 cm diameter.

Bulk density increased with a decrease in diameter. For 2.2-mm to 0.36-mm diameter aggregates this increase was due to a surface area effect, i.e., loss in porosity as a result of subdividing larger aggregates. Both surface area effects and increased porosity from root channels and fissures were required to explain the increase for clods between 5 and 1 cm diameter.

An evaporation and a sorption technique are described and used to estimate the water content of aggregates at saturation, i.e., the intra-aggregate water content. The evaporation technique gave reliable results at specific rates of evaporation, but the sorption technique was

not as useful. Clods and aggregates of the soil and of pumice samples did not fully saturate; entrapped air accounted for 25 to 27% of the total pore volume.

Aggregates of 1 and 2.2 mm diameter and clods of 1.0 and 5.0 cm diameter took about 1, 1.7, 45, and 1,000 seconds, respectively, to saturate. This rapid wetting would minimize differences in potential between inter- and intra-aggregate water during infiltration.

Additional Index Words: clay soils, swelling, infiltration.

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²Graduate Research Student and Professor, Dep. of Soil Science. The senior author is now lecturer in Soil Science, University of the West Indies, Trinidad.

TWO CLASSES of pores can be distinguished in an aggregated soil, the larger inter-aggregate pores and the much smaller intra-aggregate pores. Inter-aggregate pores are important in drainage and aeration, while intra-aggregate pores are important for retention of plant-available water and plant nutrients. Philip's theory for the prediction of infiltration into aggregated media depends on the relative diffusivities and conductivities of water in inter-

and intra-aggregate pores (Philip, 1968). The rate of wetting of the aggregates is another parameter which must be known. A slow rate of wetting creates a local disequilibrium in the profile, with water in the intra-aggregate pores under lower potential than the water in inter-aggregate pores. A third parameter which must be accurately measured is the maximum water content of the aggregate. This is determined by intra-aggregate pore volume, which may increase on wetting, and by the degree of saturation with water. Some air is normally present in the intra-aggregate voids.

The bulk density of aggregates must be known to obtain values for maximum water content at saturation. Several authors (Chepil, 1950; McIntyre and Stirk, 1954; Currie, 1966; Voorhees, Allmaras, and Larson, 1966) have proposed methods for measuring the bulk density of small aggregates. There are different views in the literature about whether bulk density of small aggregates increases with decreasing aggregate size (Wittmuss and Mazurak, 1958), or whether differences are due to surface area effects (Currie, 1966) or to inaccuracies in the techniques used in the measurements.

This study reports measurements of bulk density of aggregates of different sizes, and evaluates the contribution of surface area to the measured bulk density. An evaporation technique is proposed and used for the estimation of the maximum water content at saturation of intra-aggregate pores in small aggregates. The rate of wetting of aggregates immersed in water is measured, and is also predicted with an equation from the theory of heat conduction. These measurements, which are difficult to make on small aggregates, are required for a description of infiltration of water into aggregated soils.

MATERIALS AND METHODS

Soil Characteristics and Aggregate Preparation:

The Ormstown silty clay loam used in this study was classified as a Humic Gleysol (Baril and Mailloux, 1950), and would come in the order Inceptisol, suborder Humaquept. This soil has highly stable aggregates. The clay fraction is dominantly clay mica and chlorite, with minor amounts of quartz, feldspar, amphibole, and montmorillonite.

Soil samples taken from the 0–30 cm depth were air dried, crushed lightly with a wooden hammer and sieved through a 2.38-mm opening sieve. Aggregates > 2.38 mm diameter were then crushed in an electrically driven rotary crusher. The following size ranges of aggregates were obtained by dry sieving: 2.38–2.00 mm; 1.17–0.84 mm; 0.42–0.30 mm; referred to subsequently as sizes 1, 2, and 3, respectively, with mean aggregate diameters of 2.2, 1.0, and 0.36 mm, respectively.

Clods of various sizes were hand picked or cut to size, for use where larger units were required.

Particle size distribution was determined by the pipette method as described by Day (1965), with organic matter destroyed during the pretreatment procedure by digesting with hydrogen peroxide. Organic matter content was determined on separate subsamples by dichromate oxidation.

Bulk Density of The Aggregates

Two techniques were used to determine the bulk density of all three sizes of aggregates. A third technique was used to determine bulk density of the larger clods. All three techniques were based on Archimedes principle.

The first technique, denoted by Tx in this study, was the tech-

nique of McIntyre and Stirk (1954). A sample of about 10 cm³ of aggregates of known weight was saturated with kerosene oil, and then drained under low suction (10 cm for sizes 1 and 2, 30 cm for size 3) to remove oil from the inter-aggregate pores and outer surfaces of the aggregates. The volume was measured as the increase in volume of kerosene oil in a narrow cylinder on the addition of the sample of aggregates.

The second technique, to be referred to as Ty, was a modification of Tx, which involved weighing. A known weight of aggregates was placed in a wire basket of about 5 cm³ capacity, which was immersed in kerosene oil in a dessicator. Saturation was assured by de-airing under vacuum. The volume of the saturated aggregates was then determined from the loss in weight on displacement in kerosene.

In the third technique, Tz, clods of known dry weight were coated with paraffin wax, and the volume of the clod determined from the loss in weight by displacement in water (Blake, 1965).

The intra-aggregate porosity (n) was calculated from the bulk density (D_b) and a measured value of 2.67 g/cm³ for particle density (D_p), using the equation $n = (D_p - D_b)/D_p$.

Intra-aggregate Water Content

The large difference in size between inter-aggregate pores (macropores) and intra-aggregate pores (micropores) was utilized in two techniques to evaluate the intra-aggregate water content on saturation. It was assumed that the rate of evaporation from, or sorption into, the macropores is greater than for the micropores. A plot of water evaporated or adsorbed against time should show two distinct regions (i) free water evaporation or infiltration into macropores, and (ii) micropore water evaporation or infiltration.

In the evaporation technique, aggregates in a single layer, or individual clods, were immersed to half of their diameters in water on indented aluminum holders. After about 1 hour, the free water was siphoned off and the holders were rotated at a constant speed of about 6 rpm on a 20 cm diameter turntable. A hairdrier was used as a source of warm air. Different rates of evaporation were achieved by regulating the distance between the air vent of the hairdrier and the aggregates and/or the temperature of the air from the hairdrier. Evaporative demand was calculated by multiplying the grams of free water evaporated per square centimeter of water surface per minute by 540 cal/g (the latent heat of vaporization of water at 100°C). Free water was contained on indented aluminum holders, and the surface area of water was estimated from the weight and depth of water on the holder using a density of water of 1 g cm⁻³. The holders plus water or aggregates were weighed at intervals to calculate water lost by evaporation as a function of time. A low rate of evaporation allowed the aggregates to conduct water rapidly enough to the surface so that the system was under steady-state water flow conditions during most of the evaporation process, resulting in a delayed break in the weight loss curve. When the evaporative energy supply was too high, surface layers of water on the aggregates evaporated too quickly to allow distinction from evaporation of free water. Again the break would come at too low a water content. Therefore, there was a maximum in the water content identified by the break in the curve as the evaporative energy increased. The maximum, which should coincide with the change in evaporation of inter- to intra-aggregate pores would occur when the conductivity of water to the surface of the aggregates was marginally less than the evaporative demand.

The other method used to measure intra-aggregate water content was by sorption of water into the aggregates. The sorption apparatus consisted of a filter funnel with porous stone. The bottom was attached to a horizontal pipette, whose position could be varied a few centimeters above or below the level of the porous stone. A single layer of 2.2 mm diameter aggregates, isolated so that they were not touching, was placed on the porous stone. Then a stopcock was opened to allow water to move up through the porous stone to wet the aggregates. Five pressure heads (+2, 0, -2, -50, and -100 mm) were used in this study and measurements were made on three samples at each pressure head. The vol-

ume of water entering the aggregates was read from the pipette as a function of time.

Volume of Saturated Clods

Clods were saturated for about 2 hours with water that entered by capillary rise from shallow water tables held in dishes. The clods had about one-third of their height submerged. Clod volume was then determined by displacement in water after coating with paraffin wax (technique Tz). Molten paraffin wax was applied to a wet clod with a small paint brush. The area of the clod in contact with the dish was coated after carefully rotating the clod to expose the uncoated portion. The weight of dry soil and the water content of the clod were found by cutting the wet clod plus wax and drying for 48 hours at 40 to 45°C. This length of time was found to be adequate for drying to a constant equilibrium value. At this temperature there was negligible loss in weight of the wax, as measured on a separate sample of wax. The weight of soil was then corrected to the oven dry basis.

Rate of Saturation of Immersed Samples

Spherical clods of 2, 5, and 10 cm diameter were cut and shaped from larger clods. At zero time a sample was immersed in water which just covered its top surface. After a given time the sample was removed, wrapped momentarily in absorbent paper or allowed to drain to remove excess surface water, weighed, and oven dried. The percent saturation was then calculated using two assumptions, swelling or no swelling during wetting. The former assumption probably gives a better estimate of the percent saturation, because it is known that most of the swelling of unconfined soil clods takes place rapidly (Gumbs and Warkentin, 1972).

Predicting the Rate of Saturation of Aggregates

The entry of water into spherical aggregates can be described by the equation

$$\frac{\partial \theta}{\partial t} = D \frac{\partial^2 \theta}{\partial r^2} + \frac{2}{r} \frac{\partial \theta}{\partial r}, \quad 0 \leq r < a, t > 0 \dots [1]$$

where θ = volumetric water content, t = time, r = distance from the surface of the sphere to the wetting front, D = water diffusivity, and a = radius of the spherical aggregate. With the following initial and boundary conditions:

$$\theta_i = 0 \quad \text{when } t = 0, 0 \leq r < a \quad [2]$$

$$\theta_f = \theta_o, \text{ when } r = a, t > 0. \quad [3]$$

Equation [1] has the solution

$$\theta_t = \theta_o \left[1 + \frac{2a}{\pi r} \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \exp[-D(n^2 \pi^2/a^2)t] \sin \frac{n\pi}{a} r \right] \dots [4]$$

Table 1—Distribution of particle sizes and organic matter within aggregates of different sizes

Soil separate	Diameter limits	Aggregate		
		Size 1 2.38–2.00 mm	Size 2 1.17–0.84 mm	Size 3 0.42–0.30 mm
	μm	%		
Clay	below 2	24.2	24.7	24.4
Silt (intermediate)	2–20	27.9	26.9	27.0
Silt (USDA)	2–50	62.3	63.0	62.4
Very fine sand	50–100	5.2	5.1	4.3
Fine sand	100–250	3.6	3.2	3.7
Medium sand	250–500	1.5	1.5	5.2
Coarse sand	500–1,000	2.0	2.5	
Very coarse sand	1,000–2,000	1.1		
Organic matter		4.9	4.9	5.2

where θ_i and θ_f are initial and final water contents, θ_t is the water content at any point r in the sphere at any time t , and θ_o is the water content at saturation.

Equation [1] is analogous to heat conduction, and the solution for the initial and boundary conditions of Eq. 2 and 3 is in Carslaw and Jaeger (1963).

RESULTS AND DISCUSSION

Particle Size Distribution

There was no significant difference in particle size among the aggregates (Table 1). Sizes 1 and 2 had the same composition, and size 3 differed only, as expected, in the sand range near the size limit of these smaller aggregates.

It was, therefore, assumed that the aggregates were similar structurally, and could be expected to have similar water retention and conducting properties.

Bulk Density of The Aggregates

The pumice samples which were porous, but with a rigid framework, were used to test whether the three techniques gave the same value for bulk density. The mean bulk densities obtained by techniques Ty and Tz were not significantly different, but Tx gave a value about 7.5% higher than the mean value by technique Tz (Table 2). One of the limitations in Tx is the uncertainty in determining the suction that removes kerosene oil only from the inter-aggregate pores and outer surfaces of the aggregates. Thus, there may be greater reliability in the results from Ty and Tz than from Tx.

Both techniques Tx and Ty showed an increase in bulk density of aggregates with decreasing size (Table 2). The relationship between aggregate density and aggregate diameter for sizes 1, 2, and 3 was curvilinear (Fig. 1). The mean bulk densities from Tx and Ty were used to plot Fig. 1 since the greater reliability of Ty over Tx was not proven. Wittmuss and Mazurak (1958) obtained a similar relationship between aggregate density, measured by the method of Chepil (1950), and aggregate diameter of a silty clay loam soil (Fig. 1). The bulk densities measured in our study were, however, higher than those of Wittmuss and Mazurak.

Currie (1966), in discussing the effect of aggregate size on porosity, showed that surface area of aggregates could have a substantial effect on their bulk density. He set up a model of spherical aggregates, considered the amount of pore space which was lost on subdividing the aggregates,

Table 2—Measured bulk densities of air-dry aggregates and clods by different techniques

Material	Mean diameter of aggregate or clod	Mean bulk density by method		
		Tx*	Ty*	Tz
	mm	g cm ⁻³		
Aggregate size 1	2.2	1.570 ± 0.004	1.582 ± 0.015	
Aggregate size 4	1.0	1.652 ± 0.016	1.603 ± 0.010	
Aggregate size 3	0.36	1.787 ± 0.058	1.792 ± 0.016	
Clod†	50			1.546
Clod	10			1.558
Clod	4			1.569
Pumice	2	0.788 ± 0.026	0.740 ± 0.0215	0.733 ± 0.018

* Each value represents the mean of three measurements, given as the mean and standard deviation.

† Clods selected were visually free of holes or fractures.

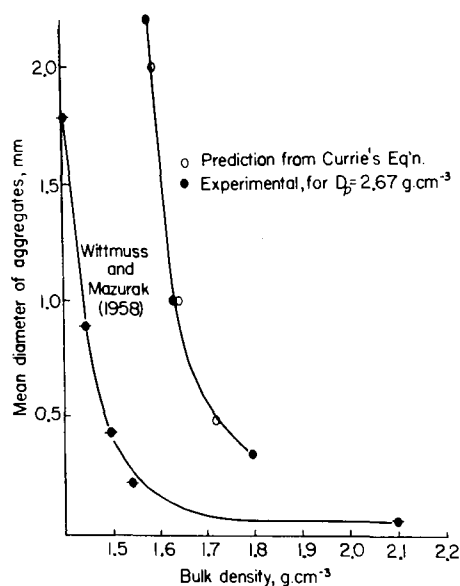


Fig. 1—Measured and predicted bulk densities of small aggregates.

and derived an expression for the effective porosity. Using this expression, the predicted changes in bulk density for a uniform material crushed to different sizes are shown in Fig. 1 and 2. Figure 1 shows the measured and predicted changes in bulk density of the 2.2 to 0.36 mm diameter aggregates used in this study. The changes in bulk density seem to be explained entirely by surface area effects. Figure 2 shows the measured values of bulk density for large clods and the changes predicted due to surface area effects. These measured changes in bulk density of the clods were not explainable as surface area effects only, but were likely a reflection of both surface area and the heterogeneity of the material, e.g., holes or fissures.

Saturated Water Content of the Aggregates

Based on a bulk density of 1.58 g/cm^3 and a particle density of 2.67 g/cm^3 , the total porosity of 2.2 mm diameter aggregates was 0.408. This was, therefore, the maximum volumetric water content of the aggregates if they did not swell. On a weight basis the water content at saturation would be 0.260 g/g. Entrapment of air would reduce the volume of pores containing water, and swelling of the aggregates would give greater porosity and higher water content. In order to estimate the percent water saturation of the aggregates (or the volume of air entrapped) from measurements of water content on a weight basis, it was necessary to know the volume increase that took place when the aggregates were saturated from the air-dry condition.

Volume increases were measured on three sizes of clods,

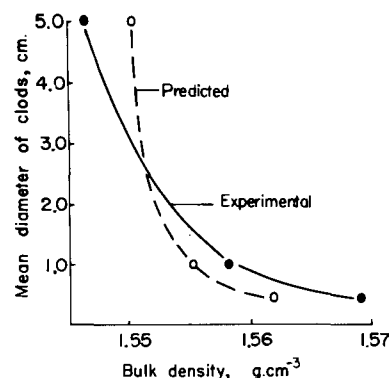


Fig. 2—Measured and predicted bulk densities of clods (prediction made using Currie's equation).

chosen at one extreme as the smallest which could conveniently be handled and at the other, as not large enough to allow the presence of visible holes or fissures. Initial dry volume was calculated from weight and bulk density measured by method Tz. Volume of saturated clods was also measured by coating with paraffin as described. Final water content of the samples, calculated volume increases, and calculated total pore volume per cm^3 and per g are given in Table 3. Saturation water content of the clods calculated from these results was $0.42 \text{ cm}^3/\text{cm}^3$ out of a total pore volume of $0.55 \text{ cm}^3/\text{cm}^3$, and therefore, about 24% of the pore volume was air filled. There were no replicates within clod sizes. However, percent volume increases were the same for all clods, and differences in water content among clods were small. It seems, therefore, that the swelling behavior of this material did not depend upon clod size in this range. In this study, it was assumed that the smaller aggregates (2.2 to 0.36 mm diameter) would swell in a manner similar to the clods.

The same methods could not be used for the aggregates because the volumes of wet aggregates could not be measured on small diameter aggregates. The evaporation technique described under "Methods" was, therefore, used to estimate saturated water content of the 2.2 mm diameter aggregates. The water content at the break in the curve of water loss versus time varied with the evaporative energy supplied (Fig. 3). At low evaporation rates, water conductance to the aggregate surface was apparently rapid enough to maintain steady water loss. At high evaporation rates, the precision of the method was not good enough to show any difference in the rate of water loss between inter-aggregate and intra-aggregate pores. Therefore, in both cases the break in the curve occurred after intra-aggregate water was already being evaporated, and the measured water content was too low. A maximum water content of 0.335 g/g was

Table 3—Final water contents, volume increases, bulk densities, and calculated saturation water contents for clods of different sizes

Air dry volume of clod	Volume of wet clod	Percent volume increase	Bulk density of wet clod*	Pore volume at saturation	Final water content of clod	Water content calculated at 100% saturation
cm ³			g/cm ³	cm ³ /cm ³	g/g	
4.06	6.33	55.9	1.217	0.544	0.350	0.447
11.96	18.64	55.9	1.207	0.548	0.349	0.454
25.20	39.25	55.9	1.190	0.554	0.358	0.466
	Mean	55.9	1.205	0.549	0.352	0.456

* Weight of dry clod/volume of wet clod.

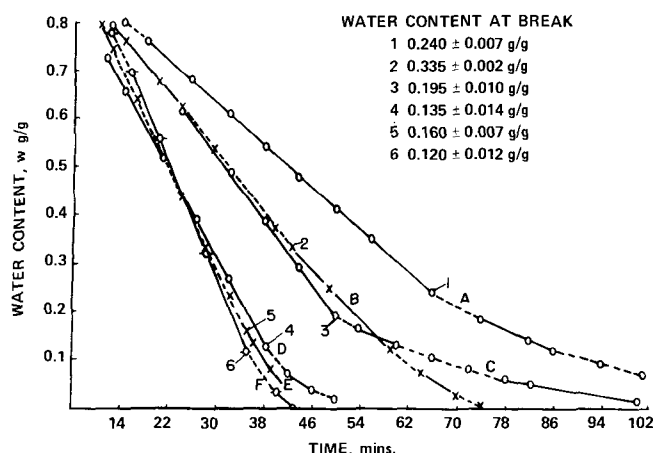


Fig. 3—Water content change during evaporative drying of 2.2 mm diameter aggregates, showing breaks (numbered 1 to 6) in curves where intra-aggregate water loss begins. The intra-aggregate water contents at these breaks are given in the inset. The evaporative demands are: A = 1.61, B = 3.13, C = 4.90, D = 8.29, E = 12.02, and F = 15.63 cal cm⁻², min⁻¹.

measured for an evaporative demand of 3.13 cal cm⁻²min⁻¹. This is the best estimate from this method for the water content of intra-aggregate pores. This water content was nearly equal to the value of 0.35 g/g measured for clods (Table 3). An evaporative demand of 3.13 cal cm⁻² minute⁻¹, therefore, appeared to be the critical value for the 2.2 mm diameter aggregates.

Measurements by the evaporation technique on pumice where swelling was not involved, and on larger soil clods where the smaller surface area to volume should reduce inaccuracies in the method, are shown in Table 4. There was a remarkably constant value of about 25% of the pores remaining air-filled, or 13% of the total soil volume. The critical evaporation energies for pumice and the clods were 3.59 and 3.13 cal cm⁻² minute⁻¹. These values were again close to the values for the smaller aggregates.

The water contents of intra-aggregate pores for isolated 2.2 mm diameter aggregates measured by the water absorp-

Table 4—Intra-aggregate water contents and percent air-filled pore space for different aggregates and clods using evaporation, sorption, and capillary rise techniques

Material	Method	Critical evaporative energy cal cm ⁻² min ⁻¹	Intra-aggregate water content g/g	Air-filled pores as % of total pore space*
Aggregates 2.2 mm diam.	Evaporation	3.13	0.335 ± 0.002	26
Pumice 4 mm diam.	Evaporation	3.59	0.610 ± 0.017	25
Pumice 10 mm diam.	Evaporation	3.59	0.62	25
Clod 1.0 cm diam.	Evaporation	3.13	0.33	27
Clod 2.0 cm diam.	Evaporation	3.13	0.34	26
Aggregates 2.2 mm diam.	Sorption 2 mm head		0.44 ± 0.042	
	0 mm head		0.42 ± 0.033	
	-2 mm head		0.41 ± 0.030	
	-50 mm head		0.31 ± 0.020	
	-100 mm head		0.27 ± 0.010	
Clod 1.0 cm diam.	Saturation by capillary rise		0.35	23
1.4 cm diam.	Saturation by capillary rise		0.35	23
1.8 cm diam.	Saturation by capillary rise		0.36	23

* Calculated on basis of the amount of swelling measured in Table 3.

tion technique for different pressures of wetting are presented in Table 4. Values given are for 50 minutes of wetting. At 2, 0, -2, and -3 mm pressure, the surface of the aggregates glistened from the presence of a film of water. The aggregates, therefore, contained more than intra-aggregate water. This excess water drained at a suction between 3 and 50 mm of water; the aggregates had a dull appearance at the latter suction. The saturated water content of intra-aggregate pores after 50 minutes wetting was, therefore, between 0.31 and 0.41 g/g, which confirmed that the value of 0.34 g/g estimated by the evaporation technique must be approximately correct.

The sorption technique could not estimate intra-aggregate water content when beds of aggregates were used because the rate of capillary movement into inter-aggregate pores was not distinctly faster than the rate of entry into intra-aggregate pores. Several rates of entry of water into the beds of aggregates were measured. The slowest rates were interpreted as water entry between the particles of the aggregates by absorption causing swelling.

Rate of Saturation of Aggregates Immersed in Water

The time required for clods of known diameter to saturate when immersed in water was measured, and an equation from the theory of heat conduction was used to predict the rate of saturation for different sizes of aggregates. The degree of saturation measured on the 2 cm diameter clods was the same for 45 and 100 seconds of immersion (Table 5), showing no increase in the degree of saturation for immersion times beyond 45 seconds. The maximum water content of 0.299 g/g was lower than 0.34 g/g measured in Table 4. This was probably due to the short time of immersion (100 seconds) which was not long enough for complete swelling.

The rate of saturation of aggregates was predicted from Eq. 4 using the following assumptions (i) the surface of the aggregate saturated instantaneously; and (ii) D was independent of water content and equal to 5×10^{-3} cm²/second, the value measured at intra-aggregate pore saturation (Gumbs, 1971).³ This latter assumption may not introduce serious errors because the diffusivities at high water contents are expected to dominate the rate of flow, particularly over the short infiltration distances for the small aggregates.

Figure 4 shows the calculated percent saturation at different fractional distances from the surface of a 2-cm diameter clod. Since Eq. 4 could be used to calculate the water content at the center of the clod, water content was calcu-

³F. A. Gumbs. 1971. Infiltration of water into aggregated soils. Ph.D. Thesis. McGill University.

Table 5—Measured and predicted rates of saturation of 2-cm diameter clods immersed in water

Duration of immersion, seconds	Percent saturation if swelling assumed*	
	Measured†	Calculated
15	41 ± 0.22	71.2
30	62 ± 0.44	89.5
45	65 ± 0.88	95.0
60	66 ± 0.44	99.6
100	65 ± 0.66	

* Total porosity if no swelling = 0.268 cm³/g; if swelling assumed = 0.456 cm³/g.

† Duplicate samples used.

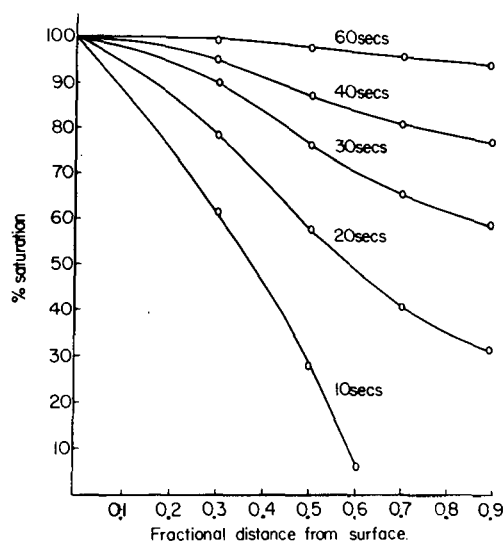


Fig. 4—Calculated wetting profiles of a 2-cm diameter clod immersed in water.

lated as a fraction of the saturation water content ($= \theta_o$) at various radial distances from the surface. The percentage saturation of the entire aggregate at any of the times given could be calculated from the water contents of a number of imaginary concentric spheres of the original aggregate. The water content of the outer spheres would contribute more to the mean water content of the aggregates than the inner spheres because of their greater volumes. The theory assumed no effects of air entrapment so that the maximum percent saturation is 100. Calculations showed that 95.0 to 99.6% saturation was achieved in 40 to 60 seconds (Table 5). The measured values in Table 5 reveal that the maximum saturation for the 2 cm diameter clod was attained after 60 seconds, and that a significant increase in percentage saturation may not have occurred after 45 seconds. The theory, therefore, estimated the correct time for saturation. For 5.0 cm clods 95 to 98% saturation was predicted in 900 to 1,200 seconds.

The theory predicted that 2.2 mm and 1 mm diameter aggregates should saturate in 2 and 1 seconds, respectively. With these rates of saturation, disequilibrium of water be-

tween inter- and intra-aggregate pores during infiltration was not likely to be serious.

Gumbs (1971)³ has shown that the rate of wetting of aggregates of Ormstown silty clay was markedly reduced if wetting took place under suction; at suctions of 4 or 5 cm of water, infiltration in media of 2.2- and 1-mm diameter aggregates took place almost entirely via intra-aggregate pores. The theory of infiltration using the diffusion equation for porous media with continuously varying pore size distribution was applicable to infiltration under suction in these media (Gumbs and Warkentin, 1975). Differences in the diffusivity of water in, or the rate of saturation of, inter- and intra-aggregate pores would, therefore, indicate how serious local disequilibrium would be during infiltration under zero or positive water pressures.

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